# Simplifying ESG evaluation for Small and Medium Enterprises

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**Abstract-This research paper explains ESG (Environmental, Social, and Governance) practices that will make the small and medium enterprises (SMEs) process easier. The primary problem is that it utilizes the approaches that are developed to meet the needs of large multinational corporations (MNCs). The purpose of this paper is to deal with the inequality that is caused by this gap. Popular schemes of evaluation suggested by big MNCs are Red-Amber-Green (RAG) ratings. This study presents the approaches that can be proposed and introduced. It will also determine the current gap with a machine learning model that will be based on accuracy, F1-score, and precision. Exploratory Data Analysis (EDA) will be used along with flow charts to measure processes and outcomes. This research has a dataset, which is grounded on international sources. The methodologies will comprise a machine learning model which will comprise of three datasets to be used in testing. Keywords: Machine Learning, ESG, Small and Medium Enterprises, Finance, Impact, Sustainability.**

***Keywords: ESG, Machine Learning , Small and medium enterprises, Finance, Impact , sustainability.***

## I. INTRODUCTION

ESG (Environmental, Social, and Governance) identifies the measures that are used to assess large MNCs. ESG is not merely about the organization, but also a way to evaluate any large organization. To illustrate, take work-life balance of employees. In the explanation of ESG, I want to explain that there is a difference between the meaning of ESG and conceptualization. We will discuss such concerns as child labour and human rights to understand whether companies comply with law and regulations. We will evaluate the effects they have on the environment and the degree of such effects. It should have threshold values; say the threshold value is 5 then a score between 0 to 5 is considered safe. A score above 5 means that there is a possibility of environmental and social damage. Environmental, Social, and Governance (ESG) framework is a way to evaluate the ethical, sustainable and societal impact of an organization. ESG has become a key element in assessing the long-term sustainability and accountability of businesses, particularly large MNCs, not only as a compliance measure, but as a comprehensive assessment tool of how organizations interact with the environment and how they treat their employees and governance. The environmental aspect pays attention to such factors as carbon emissions, energy consumption, waste disposal, and the ecological impact in general. The social aspect examines the welfare of employees, diversity and inclusion, labour practices and human rights concerns, like child labour and poor working conditions. The governance aspect audits the corporate policies, transparency, board structure, ethics and adherence to legal standards. As an example, a company with a healthy work-life balance, where labour is not exploited, and environmental regulations are observed is regarded as positive in ESG. On the other hand, firms that are engaged in environmental degradation, not transparent, and breach labour standards score low ESGs. Although the importance of ESG is increasing, the application of ESG between large MNCs and SMEs remains very wide. Majority of ESG assessment systems, tools and scoring systems such as RAG rating, are primarily designed to support large organizations with strong data, financial capabilities, and dedicated staffs. Such approaches usually do not take into account the difficulties of SMEs, including unstructured. data, financial constraints, and[1][3] low knowledge about ESG practices. This unequal situation brings about a high level of inequality that results in under-appraisal or misrepresentation of the SMEs that may be involved in sustainable practices. SMEs will not be able to participate in sustainability-driven markets and investments without simplified and flexible ESG evaluation procedures. To tackle this issue, this research proposes a simpler method for evaluating ESG for SMEs using machine learning techniques. The research aims to draw attention to the discrepancy between current ESG standards, which are developed by bigger businesses, and the real potential of SMEs. This discontinuity will be quantitatively evaluated with the help of machine learning models that assess performance gaps in terms of such metrics as accuracy, precision, and F1-score. Exploratory Data Analysis (EDA) will be used to establish trends and correlations among ESG data. Flow representations will also be used in the research to explain the data handling, model training, and gap identification. An evaluation system based on threshold will be presented to simplify the ESG scoring. To illustrate, ESG scores may be clustered in a range (e.g., 0 to 5 means the performance is safe, whereas the scores above the mark may show that there is a potential threat to the environment or the society). This arrangement helps in decision making and is easier to adopt by SMEs. The data that will be used in this study is obtained all over the world in order to have a general and homogenous perspective. Nevertheless, the methodology will be flexible and open to modifications depending on the regional or sector needs. On the whole, the study will address the ESG evaluation gap by providing a data-driven, scalable, and SME-friendly model allowing smaller companies to evaluate their sustainability performance, improve it, and report it..

## II. PROBLEM STATEMENT

The increased significance of Environmental, Social, and Governance (ESG) systems has greatly changed the evaluation of organizations in terms of sustainability and ethical impact. Nonetheless, the existing ESG assessment tools are mainly focused on big MNCs, which possess structured data platforms, financial assets, and sustainability staff. Small and Medium Enterprises (SMEs), on the contrary, face numerous problems in the implementation of these frameworks. The absence of standard data collection practice, insufficient knowledge on the concept of ESG, and affordable analytical tools are significant obstacles to SMEs. This means that smaller organizations cannot be evaluated using ESG assessment tools such as RAG ratings or the results would be inaccurate. This translates to a huge appraisal gap whereby SMEs are either undermarked or poorly gauged in sustainability measures. In addition, the models being used are not realistic in terms of real world conditions like informal labour practices, partiality in environmental standards and incomplete governance structures. The other most significant problem is that there is no quantifiable means of measuring the ESG disparity between SMEs and the set standards. Without any measurement, it would be difficult to see how far the organizations are to acceptable sustainability standards. Therefore, it requires an easier, more scalable, and data-driven model of ESG evaluation that: 1. Makes sense with the data availability of SMEs 2. Precisely determines weaknesses in ESG performance 3. Produces quantifiable outputs with machine learning III. Objectives The main objective of this research is to create a data-driven approach for analyzing ESG (Environmental, Social, and Governance) performance in Small and Medium Enterprises (SMEs) using machine learning techniques. The research aims to streamline the process of ESG assessment and overcome the shortcomings of the current models, which are largely applicable to larger corporations. The particular aims of the study are: 1. Comprehending and examining ESG parameters, such as environmental, social, and governance factors, in the framework of SMEs. 2. Gathering and refining ESG data worldwide to guarantee quality and consistency to be analyzed. 3. Performing Exploratory Data Analysis (EDA) to discover patterns, trends, and relationships between ESG indicators. 4. Calculating an overall ESG score based on environmental, social, and governance features. 5. Building a machine learning model to categorize organizations as performance types, e.g., Safe and Risk. 6. Evaluating the accuracy, precision and F1-score of the model as standard metrics. 7. Determining the ESG gap through a comparison of actual ESG performance and benchmark values. 8. Offering insights that can assist SMEs in enhancing their sustainability performance and decision-making. IV. Connected literature ESG (Environmental, Social, and Governance) research has received a lot of interest in the recent past, particularly in terms of its impact on organizational performance and sustainability. One extensive study in this field by Gunnar Friede, Timo Busch, and Alexander Bassen (2015) analysee over 2000 empirical studies to explore the link between ESG criteria and corporate financial performance. Their results indicate that almost 90 percent of these studies suggest a non-negative correlation between ESG and financial performance with a significant proportion of them recording positive outcomes. This study presents solid proof that ESG practices are advantageous to sustainability and financial growth and value creation in the long-term. More studies underline that ESG models are primarily created to be applied to large institutions, and it is difficult to apply them to SMEs. In many cases, SMEs lack structured tools, resources, and standardized approaches to effective measurement and improvement of ESG performance. This makes a significant difference between sustainability assessment and reporting of large corporations and SMEs. Also, the available literature records discrepancy in ESG scoring, and assessment procedures among various agencies and datasets. Differences in ESG ratings may also result in divergent interpretation and decision-making and it is difficult to come up with a universally recognized standard of ESG assessment.

## III. Objectives

The main goal of this research is to create a data-driven method for analyzing ESG (Environmental, Social, and Governance) performance in Small and Medium Enterprises (SMEs) using Machine Learning techniques[4][5]. The study wants to make ESG evaluation easier and tackle the shortcomings of existing frameworks that mainly cater to large organizations.

1. The particular aims of the study are as follows:
   To learn and research about the parameters of ESG that comprise environmental, social, and governance considerations within the context of SMEs..
2. To collect and pre-process ESG data of international sources to obtain quality and consistency of data to analyse.
3. To perform Exploratory Data Analysis (EDA) to find patterns, trends, and relationships between ESG indicators.
4. To compute a total ESG score of the features of the environmental, social, and governance.
5. To develop a Machine Learning model to categorize organizations into performance categories that include Safe and Risk. To provide actionable insights that can help enhance their sustainability performance and decision-making.
6. To evaluate the performance of the model in terms of conventional measures like accuracy, precision, and F1-score.
7. To determine the gap in ESG performance by making comparisons between current ESG performance and benchmark.
8. To give practical recommendations that can enable SMEs to improve their sustainability performance and decision-making.ng.

## V1. Related work

The study of ESG has gained significant interest in the recent times particularly in the context of its influence on the performance and sustainability of organizations. A comprehensive research in this area is by Gunnar Friede, Timo Busch, and Alexander Bassen (2015)[3], who examined more than 2000 empirical studies to examine the relationship between ESG criteria and corporate financial performance. Their results indicate that nearly 90 percent of the studies report non-negative relationship between ESG and financial performance with most of the studies reporting positive outcomes. This research offers substantial support to the idea that ESG practices do not only promote sustainability, but also financial growth and value creation in the long run. Additional studies reveal that ESG frameworks are primarily meant to be applied in large companies, and SMEs find it difficult to use these frameworks. SMEs do not usually have well-organized tools, resources, and standardized ways to efficiently measure and enhance their performance related to ESG. This poses a huge difference between large companies and SMEs regarding sustainability assessment and reporting. Moreover, the current literature reveals discrepancies in ESG scoring and assessment approaches of various agencies and datasets. The differences in the interpretation and decision-making due to differences in ESG ratings can make it hard to create a universal standard of ESG assessment. This problem prompts the necessity of more flexible and less complex models, especially in the case of SMEs. Recent advancements in research have also looked into the use of Machine Learning techniques in ESG analysis. Compared to conventional statistical approaches, Machine Learning models are able to detect more intricate patterns in ESG data, and make predictions more accurately. Following these approaches, it is possible to classify, assess risks, and evaluate performance on the basis of ESG indicators automatically. In these improvements, however, there are no single integrated frameworks of ESG scoring, classification, and gap analysis that are specifically designed to be applicable to SMEs. The majority of the current literature pays attention to financial performance or ESG measurement alone without considering the difference between the actual performance and the benchmark. Consequently, this study contributes to the current literature on ESG by developing a Machine Learning-based model, which in addition to assessing ESG performance can also determine the difference between the current and anticipated levels of sustainability. This will help develop a more feasible and scalable ESG assessment framework of SMEs.

## V. Methodology

The research methodology will help to analyze ESG data systematically and identify performance gaps in Small and Medium Enterprises with the help of Machine Learning. The methodology is a systematic procedure that encompasses data gathering, preprocessing, exploratory analysis, modeling and assessment..

The research starts with gathering ESG-related information on the global level, such as open sustainability data, publicly available ESG reports, financial and environmental indicators. This structured classification makes sure that all the key areas of ESG are well-represented. Preprocessing occurs after data collection to improve the quality of data and make it consistent. This includes dealing with missing values using the correct methods, eliminating duplicates, and standardizing the ESG scores to be on the same scale. Also, nominal data is transformed into numerical data to prepare it to be used in

Machine Learning models. These measures are important in preparing
the data set to be effectively analyzed.

Exploratory Data Analysis (EDA) is performed as soon as the data is prepared to get an idea of the structure and distribution of the dataset. It includes trend analysis of the ESG scores, correlation between the environmental, social, and governance variables, and identification of any outliers or other anomalies in the data. The information provided by

To simplify the process of ESG performance interpretation, a threshold-based assessment technique is proposed. In this method, ESG scores are put in specific ranges. Safe or sustainable zone is the range of scores that lie within a certain range (such as 0 to 5) and a risk zone is a range that contains scores higher than the threshold. This classification can provide a clear picture of whether an organization is functioning within reasonable sustainability parameters or is causing environmental or social problems..

The core of the methodology uses Machine Learning models[7] to analyze ESG data and predict performance. Linear Regression, Decision Trees, and Random Forest[7] models are used here. Linear Regression estimates continuous ESG scores where the Decision Trees and Random Forest models are applied in classification tasks and improved predictive power. The processed dataset is used to train these models and test their generalization to new data.

The performance of the models is measured using standard evaluation measures such as accuracy, precision and F1-score. The overall correctness of predictions is measured by accuracy, the reliability of positive predictions is measured by precision and a balanced measure of accuracy and precision is given by the F1-score. These indicators make sure that the model is capable of identifying the levels of ESG performance and gaps in an accurate manner[4].

Lastly, the ESG gap analysis is achieved through comparison of the calculated ESG scores of SMEs with benchmark values. Such a comparison aids in determining the extent to which SMEs are not as expected in terms of sustainability. The findings shed light on certain aspects to be improved and give the understanding of the sustainability level of these organizations.

*4.1 Data Collection*

1. The dataset used in this research is collected from global ESG sources to ensure diversity and reliability. These sources include:
2. Open sustainability datasets,
3. Public ESG reports, and
4. Financial and environmental indicators.
5. The dataset is further categorized into three major dimensions:
6. Environmental data, which includes factors like emissions and energy usage,
7. Social data, which focuses on laboir practices and employee welfare,
8. Governance data, which includes transparency and compliance-related indicators.

*4.2 Data Preprocessing*

1. Before applying Machine Learning models, the dataset is pre-processed to ensure quality and consistency. The preprocessing steps include:
2. Handling missing values using appropriate techniques,
3. Removing duplicate entries to avoid redundancy,
4. Normalizing ESG scores to maintain a uniform scale, and
5. Converting categorical data into numerical format for compatibility with Machine Learning models.
6. This step ensures the dataset is clean, consistent, and suitable for accurate model training.

*4.3 Exploratory Data Analysis (EDA)*

1. EDA is performed to understand the structure and distribution of the dataset. This includes:
2. Identifying trends in ESG scores,
3. Understanding relationships between environmental, social, and governance factors, and
4. Detecting outliers and anomalies in the dataset.
5. This analysis helps in selecting the most relevant features for model development and improves overall model performance.

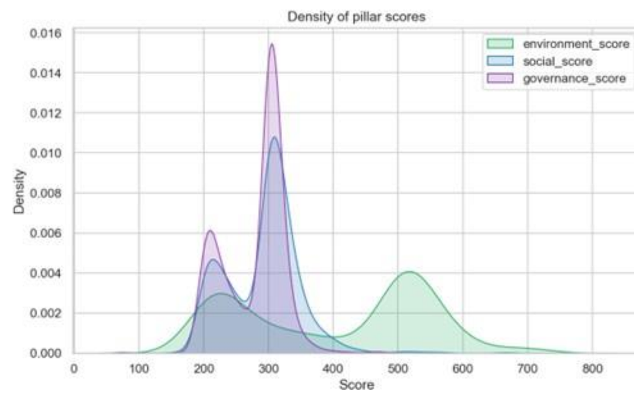


**Fig. 1.** Density distribution of ESG pillar score, environment, social, and governance showing the spread and concentration of scores across observations.

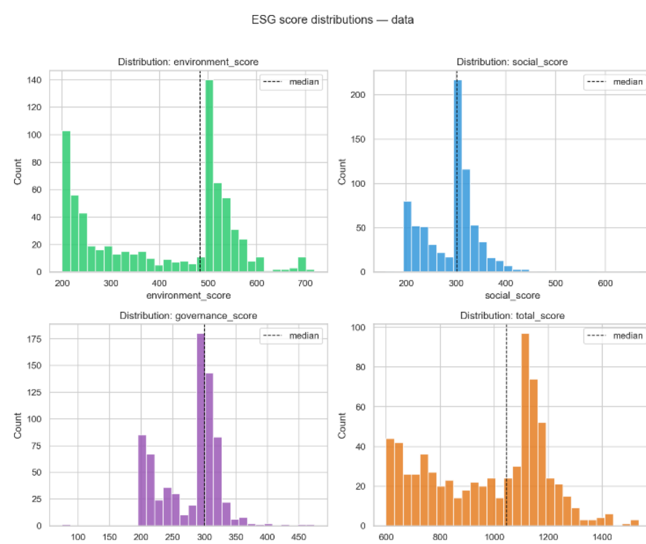


**Fig. 2.** Histograms of ESG score distributions environment, governance, and total scores with median values indicated, illustrating the frequency and spread of scores across each pillar.

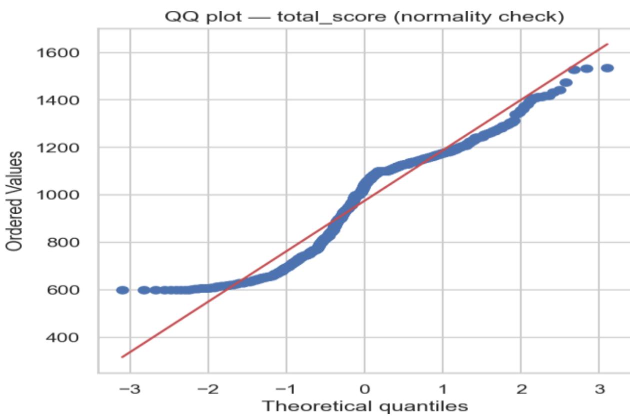


**Fig.3**. The QQ plot shows that total ESG scores deviate from normality, especially at the tails, indicating a non-normal distribution.

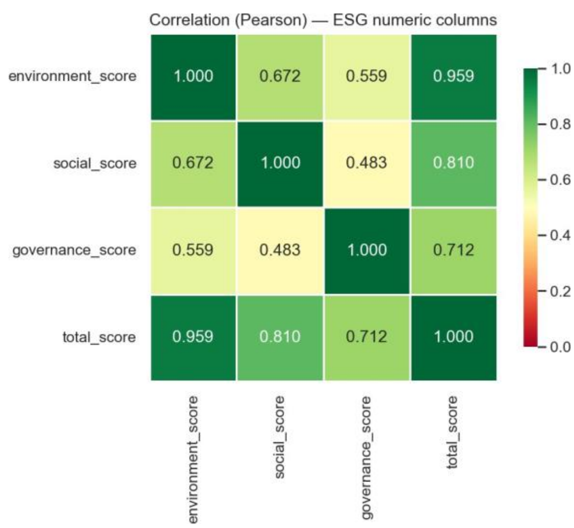


**Fig.4**.This is the correlation between the environmental score, Social and governance score.

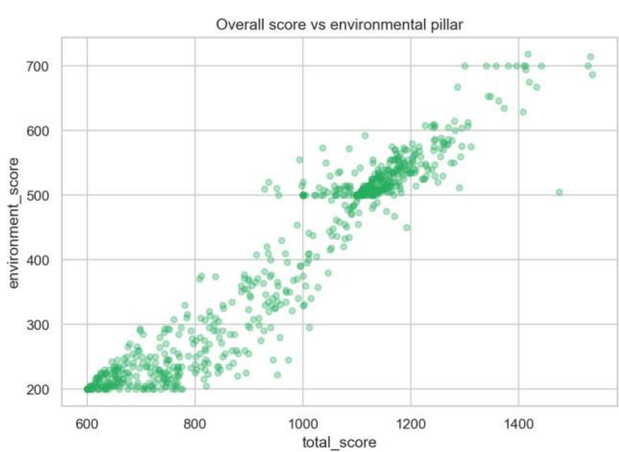


**Fig.5.** This is the environmental score on the basis of total_score. How much is the environmental score.

*4.3* Threshold-Based ESG Evaluation

1. To simplify interpretation, a threshold-based scoring system is introduced. The evaluation is defined as follows:
2. ESG scores within the range of 0 to 5 are considered to be in the safe or sustainable zone, and
3. scores above the threshold value (greater than 5) are classified as belonging to the risk zone, indicating potential environmental or social impact.
4. This approach provides a clear and interpretable mechanism to evaluate organizational performance.

*4.5. Machine Learning Model Development*

1. Machine Learning models are used to calculate the ESG gap and predict performance. The models used in this research include:
2. Linear Regression, which is applied for continuous ESG score prediction,
3. Decision Tree, which is used for classification of ESG categories, and
5. Random Forest, which enhances accuracy and robustness through ensemble learning.
6. These models are trained on the processed dataset and evaluated to ensure reliable predictions.

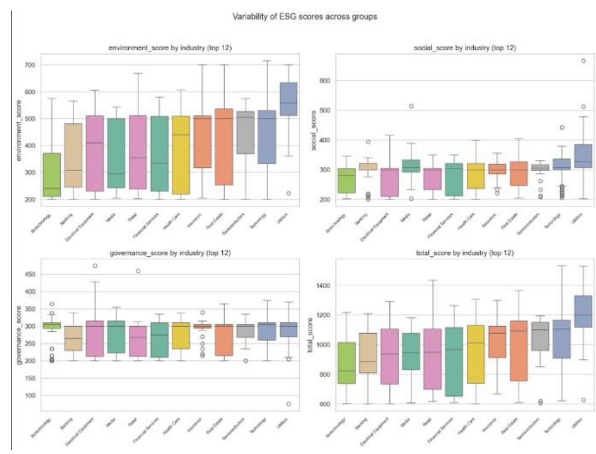


**Fig.6** Box plots showing how ESG scores (Environment, Social, Governance, Total) vary across the top 12 industries.

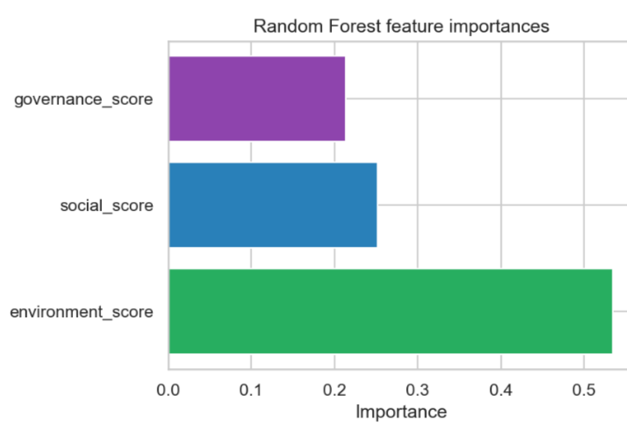


**Fig 7**. Box plots confirming Large/MNC companies consistently score higher than Small/other companies across all ESG dimensions.

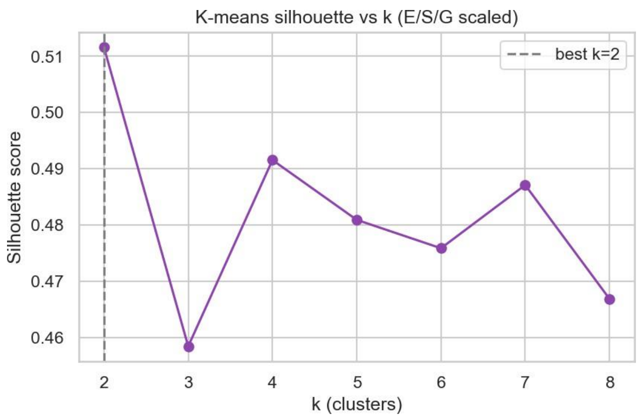


**Fig.7**. Line chart showing k=2 is the optimal number of clusters based on the highest silhouette score.

*4.6 Model Evaluation*

1. The performance of the Machine Learning models is evaluated using standard metrics. These include:
2. Accuracy, which measures the overall correctness of predictions.
3. Precision, which evaluates the correctness of positive predictions, F1-score, which provides a balance between precision and recall.

These metrics help in understanding how effectively the model identifies ESG performance and associated gaps..

*4.7 ESG Gap Analysis*

The ESG gap is calculated by comparing the actual ESG performance of SMEs with standard benchmark values.

This involves:

1. Measuring the difference between actual ESG scores and benchmark values,
2. Identifying areas where organizations fall below expected standards, and
3. Actually, evaluating the level of deviation from sustainability goals.
4. This analysis highlights key areas of improvement and provides insights into enhancing ESG performance.

**The dataset used in this research is collected from global ESG sources including**.

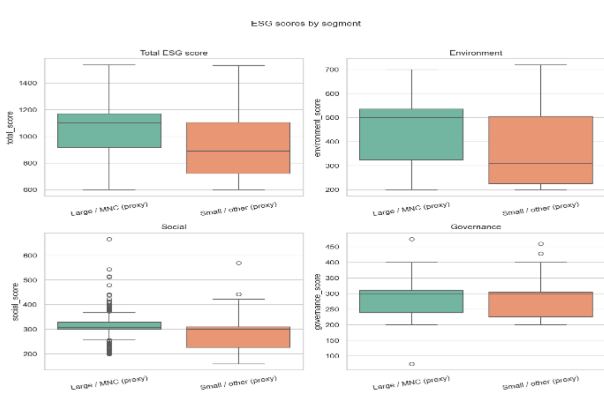


**Fig. 9.** Box plots confirming Large/MNC companies consistently score higher than Small/other companies across all ESG dimensions.

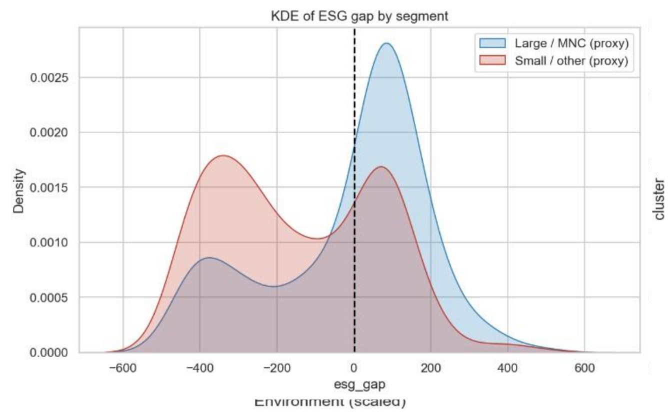


**Fig.8.** [8] KDE plot showing Large/MNC companies have a positive ESG gap while Small/other companies skew negative.

## VI. Implementation

*6.1 Tool and Technologies.*

The proposed ESG gap analysis model is implemented with Python as the main programming language because it has a high level of support of data analysis and Machine Learning. The implementation uses libraries like NumPy to perform numerical computations, Pandas to handle and manipulate data, Matplotlib and Seaborn to visualize in the course of Exploratory Data Analysis (EDA), and Scikit-learn to construct and test Machine Learning models. The model is developed and run on platforms like Vscode, Jupyter Notebook that offer an interactive environment to run the experiment and visualize.

*6.2. Dataset Implementation*

The data set in this study is ESG related indicators that are classified into three broad dimensions. Carbon emission and consumption of energy are among the environmental characteristics. The social features entail the scores of employee welfare and labour compliance like child labour and human rights. These governance characteristics are transparency scores and board structure. Everything is transformed into numbers to make it usable with the Machine Learningalgorithms.

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*6.3 Step-by-Step Implementation*

The process of implementation has a systematic series of actions to be followed to achieve proper analysis and the model performance.

1. The system loads the dataset in the first step through data handling libraries, which can be used to initially inspect and learn about the data structure.
2. Data preprocessing in the second step will involve processing the missing values, cleaning up the data and selecting the meaningful features including environmental, social, and governance scores.
3. The third step involves calculating the ESG score through weighted combination of the environmental, social, and governance factors, which guarantee that each of the three dimensions is proportionate to the aggregate score.
4. The fourth step involves a threshold-based classification of organizations classified as either safe or risky, according to their ESG score.
5. The fifth step involves Exploratory Data Analysis (EDA), where visualization (histograms) is used to learn the distribution of ESG scores.
6. The sixth step involves splitting the dataset into a training and testing subset and training a Random Forest Machine Learning model

with the chosen ESG features. In the seventh step, the trained model is analyzed at the performance metrics which include accuracy, precision, and F1-score to determine the predictive ability of the model.

7. The ESG gap is shown in the last step as the ESG score of every organization will be compared with a predetermined benchmark value and this will allow the realization of deviations in performance.

*6.4 Output of the System*

The system implemented produces several outputs which can be used in analysis and decision-making. ESG score of an individual organization, which is the overall sustainability performance. Organizations are classified as being in the Safe or Risk category. Gap value that shows difference with the benchmark. The model performance measures such as accuracy, precision, and F1-score.

| | precision | recall | f1-score | support |
|---|---|---|---|---|
| Risk | 0.96 | 0.99 | 0.97 | 78 |
| Safe | 0.98 | 0.96 | 0.97 | 67 |
| accuracy | | | 0.97 | 145 |
| macro avg | 0.97 | 0.97 | 0.97 | 145 |
| weighted avg | 0.97 | 0.97 | 0.97 | 145 |

**Fig.10**. Random Forest classification report showing 97% accuracy in predicting whether a company is "Risk" or "Safe" based on ESG scores.

*Integration with System Architecture*

The application may be expanded into a full-stack application by combining the Machine Learning model with a web-based application. The backend may be written in a Python-based framework, like Flask, with the trained model implemented as an API. The frontend, which is created on the basis of technologies like React, can offer a dashboard where users could enter ESG data and visualize the outcomes like ESG scores, classifications, and gap analysis in real-time.

## VII. Conclusion

The study introduces a Machine Learning-powered framework to analyse the ESG (Environmental, Social, and Governance) performance of Small and Medium Enterprises (SMEs). The paper identifies the current gap in the ESG assessment tools which are mostly tailored to large organizations and can not be conveniently tailored to the SMEs. The proposed model combines the efforts of Exploratory Data Analysis and Machine Learning to offer a systematic way of assessing the ESG performance. The classification of organizations as Safe and Risk could be performed with the help of the Rand Forest classifier, and the gap analysis of the ESG could help reveal the inconsistency with the traditional standards of sustainability. The outcomes show that the model is highly accurate and is able to reflect trends in ESG data, which is why it is an effective instrument in sustainability assessment. The framework does not only make ESG assessment easy but also offers practical lessons that can help SMEs make decisions and policy formulations. All in all, the study will help fill the gap between the old approaches to ESG assessment and the concrete requirements of SMEs by offering a scalable, data-driven, and interpretable solution.

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